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Depth gauge seal

Abstract

A seal for a depth gauge having an axis of revolution, the seal including a radially inner flank and a radially outer flank, wherein the radially inner flank has a height (h1) greater than a height (h2) of the radially outer flank. Also, a depth gauge with the seal.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is claiming priority based on European Patent Application No. 21171306.0 filed on Apr. 29, 2021, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD OF THE INVENTION

(2) The field of the invention relates to devices for sealing a depth gauge subjected to the pressure of a fluid.

(3) The invention relates more particularly to a seal of a depth gauge, and more particularly of a mechanical depth gauge, comprising a membrane pressure sensor, the seal ensuring tightness at the membrane pressure sensor.

(4) The invention also relates to a timepiece comprising such a mechanical depth gauge.

TECHNOLOGICAL BACKGROUND

(5) As illustrated in FIG. 1, mechanical depth gauges **1** make it possible to indicate water depth, for example by means of a rotating hand **2** facing a graduation **3** of a dial **4**, particularly using a pressure sensor housed in the depth gauge case **6**. The pressure sensor is conventionally connected to the hand **2** by a mechanical transmission.

(6) The pressure sensor is generally housed in a pressure chamber arranged in the case **6**, such a chamber is intended to receive a pressurised fluid.

(7) In mechanical depth gauges, it is common for the pressure sensor to comprise a deformable membrane, for example in the form of a metallic disk comprising concentric undulations so as to modify the amplitude of elastic deflection.

(8) To fasten such a pressure sensor and ensure tightness between the inside and the outside of the depth gauge case, it is known to solder the deformable membrane, see for example patent application DE 10 147 124, or to grip the deformable membrane as proposed in the document WO 01/01098. However, the manufacture of these deformable membranes is relatively complex and does not have a satisfactory reproducibility. Moreover, it is not easy to ensure that the deformable membrane will not enter the plastic deformation range thereof if the pressure sensor were subjected to a pressure exceeding the service pressure thereof.

(9) To remedy these drawbacks, manufacturers have replaced these metallic disks with undulations by flat deformable membranes, but the fastening method continues to create quite substantial drawbacks. Indeed, if the peripheral region of the deformable membrane is soldered to the sensor structure, this reduces the elastic deformation that the membrane can withstand before being deformed plastically, therefore the sensitivity of the pressure sensor is reduced.

(10) Moreover, soldering introduces different stiffness characteristics for each solder. The resulting inaccuracies on the deflections of the deformable membrane obviously reduce the precision of the sensor and, furthermore, make it difficult to use bankings preventing plastic deformations of the deformable membrane. Fastening the deformable membrane by setting in the structure of the sensor also has some of the drawbacks mentioned above.

(11) To partially remedy these drawbacks, a pressure sensor of simple design has been proposed, enabling the deformable membrane to flex as freely as possible under the pressure of the fluid while retaining satisfactory tightness thanks to the use of a seal.

(12) Thus, the document EP 2 264 399 proposes not rigidly connecting the peripheral region of the deformable membrane to the body of the depth gauge, or to the cover, such that the peripheral region of the deformable membrane can pivot on a banking strip when the deformable membrane flexes under the effect of an increase in pressure of the fluid in the pressure chamber. Such an

architecture is represented by FIG. 2, which illustrates a peripheral region of the case 6 of a depth gauge 1.

(13) To ensure the tightness of such an assembly, an O-ring type seal 7 is compressed against the peripheral region of the deformable membrane 8 to ensure the tightness of the inner volume of the depth gauge in relation to the pressure chamber 10. The seal 7 furthermore serves as a support element to press the peripheral region of the deformable membrane 8 constantly against a banking 9, particularly when the deformable membrane 8 pivots.

(14) The O-ring 7 is housed in a circular groove 11, with a rectangular profile, formed at the bottom of the depth gauge. Such a circular groove 11 has the advantage of facilitating the positioning of the O-ring 7 during assembly. Such a circular groove 11 also makes it possible to limit the radial spreading of the O-ring 7 upon the compression thereof, during the assembly as well as the pressurization of the pressure chamber 10 and therefore the deformation of the deformable membrane 8. The circular groove 11 therefore makes it possible to ensure a sufficient contact pressure on the deformable membrane 8 to ensure the required tightness in the range of use of the depth gauge.

(15) However, in the case of intensive use of the depth gauge, as described in the application EP 2 264 399 and illustrated in FIG. 2, and/or in the case of poor cleaning thereof by the user, salt residue can build up in the circular groove 11 or in the vicinity of this circular groove, resulting in a risk of corrosion of this region and in premature wear of the deformable membrane 8.

SUMMARY OF THE INVENTION

(16) In this context, the invention proposes a novel depth gauge seal having a profile particularly suitable for use at a pressure sensor, particularly with a deformable membrane making it possible to correctly ensure the required tightness levels for such a use, while making it possible to simplify the geometry of the different elements at the sealing zone of the depth gauge so as to prevent the build-up of salt residues in this peripheral region of the deformable membrane.

(17) For this purpose, the invention relates to a seal for a depth gauge of axis of revolution Z comprising a radially inner flank and a radially outer flank, characterised in that the radially inner flank of said seal has a greater height than a height of the radially outer flank.

(18) Besides the features mentioned in the paragraph above, the seal according to the invention can have one or more of the following additional features, considered individually or according to any technically possible combinations: the seal has a section, along a radial cutting plane, having a polygonal shape; the section has a trapezoidal shape comprising two parallel sides, the two parallel sides of the trapezium being formed by the radially outer flank and the radially inner flank; the section has a top face comprising at least one inclined planar portion with respect to a bottom face; the top face comprises a first planar portion parallel with a bottom face and an inclined planar portion with respect to the bottom face; said at least one inclined planar portion has an inclination with respect to said bottom face, said inclination being greater than 0° and less than or equal to 45°; the seal has a section, along a radial cutting plane, having a quarter-circle shape; the seal is made of elastomer, preferably of nitrile; the seal is configured to be compressed against a deformable membrane of a depth gauge pressure sensor.

(19) The invention also relates to a depth gauge comprising: a case closed by a bottom, the bottom delimiting a pressure chamber in fluidic communication with the outside of the depth gauge, a pressure sensor comprising a deformable membrane disposed above said pressure chamber; a seal according to the invention, the seal ensuring the tightness of said case in relation to said pressure chamber.

(20) Advantageously, the seal is compressed and forms a support element of said deformable membrane.

(21) Advantageously, the seal is disposed at a peripheral region of said deformable membrane.

(22) Advantageously, the peripheral region of the deformable membrane can pivot pressing on the seal when the deformable membrane is deformed under the effect of an increase in pressure in the

pressure chamber.

(23) Advantageously, the bottom is devoid of a channel or a groove for housing the seal.

(24) Advantageously, the bottom comprises a top face delimiting a bottom portion of said pressure chamber, the seal being positioned directly on the top face of the bottom delimiting said bottom portion of said pressure chamber.

(25) Advantageously, the depth gauge comprises an indexing member provided on the top face of the bottom to facilitate the centering of said seal.

(26) Advantageously, the seal delimits a peripheral portion of said pressure chamber.

(27) Advantageously, the deformable membrane is formed by a metallic disk or by an amorphous metal alloy disk.

(28) Advantageously, the disk is flat.

(29) The invention also relates to a timepiece, for example a watch, and more particularly a diving watch, comprising a depth gauge according to the invention.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The aims, advantages and features of the present invention will emerge on reading the following detailed description with reference to the following figures.

(2) FIG. 1 is a schematic representation of a front view of a depth gauge equipped with a membrane pressure sensor according to the prior art;

(3) FIG. 2 is a partial radial sectional representation along the line II banking's of the peripheral region of the depth gauge according to the prior art illustrated in FIG. 1;

(4) FIG. 3 is a schematic and partial radial section view of an embodiment example of a depth gauge according to the invention equipped with a membrane pressure sensor and a seal according to the invention;

(5) FIG. 4 is an enlarged view of detail III in FIG. 3, illustrating more particularly the peripheral region of the pressure sensor and the seal according to the invention when the depth gauge is not subjected to external pressure greater than atmospheric pressure;

(6) FIG. 5 is an enlarged view of detail III in FIG. 3, illustrating more particularly the peripheral region of the pressure sensor and the seal according to the invention when the depth gauge is subjected to external pressure greater than atmospheric pressure;

(7) FIG. 6 is a schematic representation of the section along a radial cutting plane of a first embodiment example of the seal according to the invention;

(8) FIG. 7 is a schematic representation of the section along a radial cutting plane of a second embodiment example of the seal according to the invention;

(9) FIG. 8 is an alternative embodiment of the first embodiment example of the seal according to the invention illustrated in FIG. 6.

(10) In all the figures, common elements bear the same reference numbers unless specified otherwise.

DETAILED DESCRIPTION OF THE INVENTION

(11) FIGS. 1 and 2 have been described above in the preamble of the invention.

(12) FIGS. 3 to 5 schematically represent a depth gauge **100**, of axis of revolution Z, according to the invention.

(13) The depth gauge **100** is, for example, a depth gauge intended to be worn on a user's wrist, thanks to a wristlet (not shown). Conventionally, the depth gauge **100** according to the invention comprises a hand **2** indicating the water depth, the hand **2** rotating facing a graduation **3** provided on a dial **4**, using a pressure sensor **17** housed in the case **6** of the depth gauge **100**.

(14) The pressure sensor **17**, positioned in the bottom part of the case **6**, is connected to the hand **2**

by a mechanical transmission member **19** configured to convert the axial movement of the pressure sensor **17**, along the axis of revolution Z of the depth gauge **100**, into a rotational movement about the axis of revolution Z.

(15) The transmission member **19** particularly comprises a rotary shaft **22** equipped with a feeler-spindle **23**. Obviously, other mechanical transmission member architectures can be envisaged without leaving the context of the invention.

(16) It will be noted that the depth gauge **100** can be combined with a watch in the same case **6**, particularly a diving watch, without however being essential. In this case, the case **6** further comprises a horological movement (not shown) and the dial **3** has graduations to indicate information relating to time information.

(17) The case **6** is closed by a pierced bottom **11**. The pressure sensor **17** is for example mounted on the bottom **11** of the case **6**, as illustrated in FIGS. **3** to **5**.

(18) The pressure sensor **17** is disposed above a pressure chamber **29** delimited by the bottom **28**. The pressure chamber **29** communicates with the outside of the case **6** through holes **30**, such that the pressure chamber **29** is in fluidic communication with the outside of the depth gauge **100**. Thus, the fluid contained in this pressure chamber **29** is subjected to the pressure prevailing around the depth gauge **100**.

(19) The pressure sensor **17** comprises a deformable membrane **12**, for example of flat and circular shape, in which the peripheral region **13** is clamped between the bottom **28** and a rigid banking **14** fastened in the case **6** capable of limiting the deformation of the deformable membrane **12** of the pressure sensor **17**.

(20) In the embodiment example represented in FIG. **3**, the rigid banking **14** is equipped with a central opening **15**, particularly for inserting the feeler-spindle **23**. Thus, the feeler-spindle **23** can move vertically in the opening **15** and pivot the shaft **22** according to the displacements (deformations) of the deformable membrane **12**.

(21) Obviously, the shape of the rigid banking **14** can be different and not have a central opening **15** according to the architecture of the transmission member used.

(22) A return spring (not shown) keeps the free end of the feeler-spindle **23** pressing against the deformable membrane **12** so as to ensure permanent contact between these two elements.

(23) Between the opening **15** and the peripheral region **13** of the deformable membrane **12**, the rigid banking **14** has a slightly concave bottom face which delimits, with the deformable membrane **12**, a chamber **18** communicating with the rest of the inner volume of the case **6** via the opening **15**. This concave face forms a stop surface **20** to limit the deflection of the deformable membrane **12** once the latter is subjected to a differential pressure exceeding a limit pressure which will be defined hereinafter.

(24) The inner volume of the case **6** contains air or another gas, at a reference pressure.

(25) A seal **21**, of axis of revolution Z, is compressed against the deformable membrane **12**, more particularly at the peripheral region **13** thereof, and ensures the tightness of the inner volume of the case **6** in relation to the pressure chamber **29** and the outside of the case **6**.

(26) The seal **21** also forms a support element of the deformable membrane **12**, configured to press the peripheral region **13** of the deformable member **12** constantly against the rigid banking **14**, regardless of the deformation state of the deformable membrane **12**. The compression of the seal **21** against the deformable membrane makes it possible to obtain a sufficient contact pressure to ensure the tightness of the case **6**.

(27) As illustrated in FIG. **5**, which illustrates more specifically the peripheral region of the pressure sensor **17** and the seal **21** according to the invention when the deformable membrane **12** is deformed and the pressure sensor **17** is subjected to a differential pressure between the chambers **29** and **18**, the seal **21** is configured to enable the pivoting of the peripheral region **13** of the deformable membrane **12** while ensuring the contact pressures required to ensure the tightness of the inner volume of the case **6**.

(28) The bottom **28** has a vertical rim **24**, arranged in a peripheral region of the bottom **28**, which abuts against the rigid banking **14** and the height of which is selected in such a way that the seal **21** is, as illustrated in FIG. **4**, substantially pre-stressed against the deformable membrane **12**, to guarantee the tightness therebetween in the entire range of service pressures of the pressure sensor **17** of the depth gauge **100**.

(29) The compression of the seal **21** presses the peripheral region **13** of the deformable membrane **12** against a part of the rigid banking **14**, that is to say a support strip **25** which follows the edge of the concave stop surface **20** and is located practically facing the position of the seal **21**. The support strip **25** is planar in this example, but it could also have a rounded or ridged transverse profile.

(30) The material and the thickness of the deformable membrane **12** are selected in such a way that the deformations of the deformable membrane **12** remain within the elastic range in the entire range of pressures to which the sensor **17** will be subjected.

(31) The deformable membrane **12** is produced from a material conventionally used and known to a person skilled in the art for such an application.

(32) Preferably, the deformable membrane **12** is a metallic disk or a disk made of amorphous metal alloy also known as metallic glass.

(33) Thanks to the planar shape thereof at rest, the method for manufacturing the deformable membrane is carried out easily, for example by cutting in a stainless steel sheet. The other elements of the pressure sensor **17**, apart from the seal **21**, can be made of metal or of rigid synthetic material, for example.

(34) When the depth gauge **100** is plunged to a certain depth in water, the deformable membrane **12** flexes elastically under the increased differential pressure between the chambers **29** and **18** located on either side of the deformable membrane **12**. The deflection of the deformable membrane **12** is transmitted to the feeler-spindle **23**. The mechanical transmission by the transmission member **19** between the feeler-spindle **23** and the hand **2** is arranged to produce a practically linear movement of the hand **2** according to the pressure variation. However, it is also possible to produce a non-linear movement of the hand **2** according to the transmission member **19** and the different geartrains used. In this case, the dial will have a non-linear graduation **3**.

(35) The shape given to the stop surface **20** corresponds advantageously to the profile of the deformed shape of the deformable membrane **12** for the limit pressure mentioned above. This profile, theoretically parabolic for a circular membrane in which the deflections are small, can be approached by a spherical dome shape, easy to machine. Preferably, said limit pressure is slightly greater than the maximum service pressure of the depth gauge **100**. As the latter must generally be subjected to a maximum test pressure substantially greater than the maximum service pressure, the main role of the stop surface **20** is that of preventing plastic deformations of the deformable membrane **12** under these test conditions, since the deformable membrane **12** is then shouldered by the rigid banking **14** which is much more rigid than it.

(36) In the embodiment example illustrated in FIGS. **3** to **5**, only the central part of the deformable membrane **12** facing the opening **15** might be subjected to additional deflection, but the additional stress is reduced and, with suitable dimensioning, can remain within the elastic range. Obviously, these advantages also exist in cases where the pressure sensor was accidentally subjected to excessive pressure, for example a water hammer in a pressure gauge.

(37) Given that the peripheral region **13** of the deformable membrane **12** is neither soldered nor set in the structure supporting it, it can pivot virtually freely on the support strip **25** to tilt and approach the banking surface **20**.

(38) It will also be noted that a countersink **26** is arranged between the seal **21** and the vertical rim **24** to enable the edge of the deformable membrane **12** to lower freely during the deformation thereof.

(39) FIG. **6** represents a first embodiment example of the seal **21** according to the invention and FIG. **7** represents a second embodiment example of the seal **121** according to the invention.

(40) FIG. 6 illustrates more specifically, schematically, the section S, along a radial cutting plane, of the first embodiment example of the seal **21** according to the invention, in the non-compressed state thereof.

(41) FIG. 7 illustrates more specifically, schematically, the section S', along a radial cutting plane, of the second embodiment example of the seal **121** according to the invention, in the non-compressed state thereof.

(42) Unlike the seals known from the prior art described in the preamble of the invention with reference to FIG. 2, the seal **21**, **121** according to the invention has a non-circular section and has no symmetry between a radially inner portion P_i and a radially outer portion P_e .

(43) In FIGS. 6 and 7, a radially inner portion P_i is positioned to the left of a dotted-line axis representing the middle of the width L of the section S, S' of the seal **21**, **121**, and a radially outer portion P_e is positioned to the right of the dotted-line axis in question.

(44) The expressions “radially inner” and “radially outer” are obviously to be taken into consideration in relation to the axis of revolution Z of the depth gauge **100** and the seal **21**, **121**, both axes of revolution being merged. Thus, a radially inner portion, flank or face is radially closer to the axis of revolution Z than a radially outer portion, flank or face.

(45) The term flank denotes the lateral part of the seal **21**, **121**.

(46) As represented in FIGS. 6 to 8, the seal **21**, **121** has, in the non-compressed state thereof, a radially inner flank **213**, **223** having a height h_1 greater than the height h_2 of the radially outer flank, therefore inducing a difference in material thickness between the radially inner portion P_i and the radially outer portion P_e of the seal **21**, **121**.

(47) For example, the ratio between the height h_1 of the radially inner flank **213**, **223** to the height h_2 of the radially outer flank **214**, **224** is between 2 and 1.25.

(48) For example, the height h_1 of the radially inner flank **213**, **223** is identical to the width L of the seal **21**, **121**.

(49) The difference in height between the radially inner flank **213**, **223** and the radially outer flank **214**, **224** makes it possible advantageously to obtain a difference in material volume, between the radially inner portion P_i and the radially outer portion P_e of the seal **21**, **121**.

(50) This difference between the radially inner portion P_i and the radially outer portion P_e makes it possible to retain a sufficient contact pressure over the entire contact surface area of the seal **21** when the deformable membrane **12** is deformed and pivots at the seal **21**, **121**, thus releasing the contact pressure previous obtained during the compression of the seal **21**, **121**, during the clamping and mounting of the bottom **28** on the case **6**. Indeed, when the deformable membrane **12** is deformed under a positive pressure, it will move towards the rigid banking **14** and therefore away from the case bottom **28**.

(51) The seal **21**, **121** according to the invention has a geometry making it possible advantageously to remove the use of a channel or a groove for housing the seal, arranged conventionally at the bottom as shown in FIG. 2, and therefore the need to have a support surface at the radially inner portion thereof in order to stress the seal radially with the aim of limiting the radial spreading thereof and maintaining sufficient compression during the deformation of the deformable membrane **12** in order to ensure the required tightness levels.

(52) Indeed, the seal **21**, **121** according to the invention has a sufficient rigidity, particularly at the radially inner portion P_i thereof, so as not to need a support surface capable of limiting the radial spreading thereof during the compression thereof against the deformable membrane **12**.

(53) Thus, the seal **21**, **121** according to the invention makes it possible to ensure the tightness of the case **6** within the entire range of service pressures of the pressure sensor **5** of the depth gauge **100**, while enabling the overpressure of such a channel or groove causing the build-up of salt residues in this region close to the peripheral region **13** of the deformable membrane **12**.

(54) Thus, the seal **21**, **121** according to the invention makes it possible advantageously to be able to remove any rib, roughness promoting a build-up of salt residues in the event of poor cleaning of

the depth gauge **100**, while ensuring the required tightness levels within the range of use of the depth gauge **100**.

(55) The seal **21**, **121** according to the invention is advantageously positioned directly on a top face **28a** of the bottom **28**, delimiting the bottom portion of the pressure chamber **29**, the top portion of the pressure chamber **29** being delimited by the bottom face of the deformable membrane **12**.

(56) With the seal **21**, **121** according to the invention, the peripheral portion of the pressure chamber **29** is delimited directly by the seal **21**, **121**, and particularly by the radially inner flank **213**, **223** thereof.

(57) As illustrated in FIGS. **4** and **5**, the bottom **28** can however have in a radially outer region in relation to the seal **21**, **121**, a wall **27**, for example vertical or slightly inclined, capable of forming a lateral banking for positioning the seal **21**, **121**. Thus, sliding thereof during assembly is prevented and the correct positioning of the seal **21**, **121** facing the support strip **25** of the rigid banking **14** is ensured. This region not being in communication with the fluid circulating in the pressure chamber **29**, there is no risk of build-up of salt residues and therefore no risk of premature degradation of the deformable membrane **12**.

(58) With reference to FIG. **6**, the first embodiment example of the seal **21** has a section S of polygonal shape, and more particularly of trapezoidal shape comprising two parallel sides, the two parallel sides of the trapezium being formed by the radially outer flank **214** and the radially inner flank **213** of the seal.

(59) The section S of the seal **21** has planar surfaces separated from one another by rounded sectors forming coupling portions between the different planar surfaces or faces.

(60) More specifically, the seal **21** comprises four planar faces: a bottom face **211**; a top face **212**; a radially inner lateral face forming the radially inner flank **213** mentioned above, and a radially outer lateral face forming the radially outer flank **214** mentioned above.

(61) The rounded sectors connecting the different faces having a section shaped in the arc of a circle, the rounded surfaces being capable of having relatively large radii of curvature.

(62) Preferably, the radially inner lateral face and the radially outer lateral face extend along a direction substantially perpendicular to the bottom face **211**. Thus, the flanks **213**, **214** of the seal **21** extend substantially perpendicularly to the bottom face **211**.

(63) The top face **212** of the seal **21** has at least one portion having an inclination with respect to the bottom face **211** for example an inclination greater than 0° and less than or equal to 45° .

(64) In the embodiment example illustrated in FIG. **6**, the entire top face **212** is inclined, however, according to an alternative embodiment illustrated in FIG. **8**, it is also envisaged that the top face **212** has a first planar portion **212a** parallel with the bottom face **211** and a second planar portion **212b** having an inclination with respect to the bottom face **211**. In this case, the inclined planar portion **212b** will be positioned in a radially outer region with respect to the first planar portion **212a**.

(65) The inclination of at least one portion of the top face **212** of the seal **21**, intended to be in contact with the deformable membrane **12**, makes it possible to improve the behaviour of the seal **21** further during the pivoting of the deformable membrane **12** as explained above.

(66) The inclination of at least one portion of the top face **212** of the seal **21** also makes it possible to ensure an optimal and more homogeneous contact pressure, during the deformation of the deformable membrane **12**, regardless of the pressure prevailing in the chamber **29**.

(67) Producing at least one inclined top surface makes it possible to ensure a sufficient contact pressure level between the seal **21** and the deformable membrane **12**, within the entire pressure range of use of the depth gauge **100**.

(68) Thus, thanks to this specific geometry, the seal **21** according to the invention makes it possible to compensate for pressure losses during the pivoting of the deformable membrane **12** due to the removal of the support walls formed by a channel or a groove. Thus, the seal **21** makes it possible to ensure the required tightness levels particularly when the membrane is deformed, as illustrated

by FIG. 5, when the tightness needs are greatest, and even in the absence of a channel or a groove limiting the spreading of the seal.

(69) Furthermore, thanks to the seal according to the invention, the contact pressures at the deformable membrane **12** and at the bottom **28** are more homogeneous over the entire surface areas in comparison with the circular O-ring in the absence of a channel or a groove.

(70) FIG. 7 illustrates a second embodiment example of the seal **121** according to the invention.

(71) The seal **121** is identical to the first embodiment example described above, except for that which will be described hereinafter. Thus, all of the features described with reference to the first embodiment example also apply for this second embodiment example.

(72) The seal **121** also comprises four planar faces: a bottom face **221**; a top face **222**; a radially inner lateral face forming the radially inner flank **223** mentioned above, and a radially outer lateral face forming the radially outer flank **224** mentioned above.

(73) In this second embodiment example, the inclined top face **212** is replaced by a planar top face **222**, substantially parallel with the bottom face **221**, and of reduced dimension (for example of the order of $L/2$) and the coupling sector **225** between the top face **222** and the radially outer face **224** has a greater radius of curvature, typically of the order of $L/2$.

(74) Thus, the top face **222** and the radially outer face forming the radially outer flank **224** have a smaller dimension than the first embodiment example, typically of the order of $L/2$.

(75) The section of the seal **121** according to this second embodiment example could resemble a quarter-circle shape.

(76) In the same way as for the first embodiment example, this shape of the seal **121**, according to this second embodiment example, makes it possible advantageously to do away with the use of a channel or a groove for housing the seal, arranged conventionally at the bottom **28**, and the need to have a support surface at the radially inner portion thereof in order to stress the seal radially with the aim of limiting the radial spreading thereof and maintaining a target compression during the compression of the seal in order to ensure the required tightness levels.

(77) The rounded surface **225** having a significant surface area also makes it possible to improve the behaviour of the seal **121** during the pivoting of the deformable membrane **12** in order to ensure an optimum and homogeneous contact pressure over an extended surface area.

(78) Obviously, other seal section profiles are envisaged without leaving the context of the invention once they have a greater material volume at a radially inner portion in relation to a radially outer portion and the top face of the seal has a profile making it possible to follow the pivoting of the deformable membrane **12** when it is subjected to a pressure differential with no loss of homogeneity at the contact pressures.

(79) The seal **21**, **121** is made of elastomer, preferably of nitrile.

(80) The bottom **28** can also comprise an indexing element **31** arranged on at least a portion of the top face **28a** of the bottom **28** to facilitate the positioning and centering of the seal **21**, **121** in the absence of a channel or a groove.

(81) This indexing element **31** forms a protuberance, such as a boss. The indexing element **31** extends circularly over at least a portion of the top face **28a** of the bottom **28**. Preferably, the indexing element **31** is circular.

(82) Such an indexing element **31** forms a visual and/or sensory indicator facilitating the positioning and centering of the seal **21**, **121** during assembly. It will be noted that this indexing element **31** has a height substantially less than the height of the seal **21**, **121** or than the vertical rim **27**. Indeed, the indexing element **31**, as described and represented, is not intended to form a channel or a groove for positioning the seal **21**, **121** but forms a slight boss capable of indicating the correct centering of the seal **21**, **121**.

(83) In respect of the assembly of the pressure sensor **17**, it is easily understood that it is particularly simple, particularly as it is simply necessary to: insert the rigid banking **14** into the case **6**, position the seal **21**, **121** on the top face **28a** of the bottom **28**, fit the sealing membrane **12** on

the seal **21**, **121**, fasten the bottom **28** to the case **6** in the usual manner; the height of the second vertical rim **24** automatically determines the pre-stress force applied to the seal **21**, **121**.

(84) The invention also relates to a timepiece, such as a watch, and particular a diving watch, comprising a depth gauge as well as a seal according to the invention capable of ensuring the tightness of the pressure sensor in relation to the timepiece case.

(85) The seal according to the invention also enables the following advantages: ensuring the required tightness levels regardless of the diving depths and therefore regardless of the positions of the pressure sensor; a similar size to the seals conventionally used in depth gauges, and particularly diving watches; removing the problems of corrosion of the peripheral region of the deformable membrane by removing the channel or the groove for positioning the seal.

Claims

1. A depth gauge comprising: a case closed by a bottom, said bottom delimiting a pressure chamber in fluidic communication with the outside of the depth gauge, a pressure sensor comprising a deformable membrane disposed above said pressure chamber; and a seal including a radially inner flank and a radially outer flank, the radially inner flank having a height greater than a height of the radially outer flank in an un-deformed state, said seal being sandwiched between the deformable membrane and the bottom to ensure tightness of said case in relation to said pressure chamber including when the deformable membrane is in a deformed state, wherein the bottom and the deformable member are devoid of a channel or a groove for housing said seal such that said seal is not retained in any channel or groove.
 2. The depth gauge according to claim 1, wherein said seal is compressed and forms a support element of said deformable membrane.
 3. The depth gauge according to claim 2, wherein said seal is disposed at a peripheral region of said deformable membrane.
 4. The depth gauge according to claim 3, wherein said peripheral region of the deformable membrane can pivot pressing on the seal when the deformable membrane is deformed under the effect of an increase in pressure in the pressure chamber.
 5. The depth gauge according to claim 1, wherein said bottom comprises a top face delimiting a bottom portion of said pressure chamber, said seal being positioned directly on the top face of the bottom delimiting said bottom portion of said pressure chamber.
 6. The depth gauge according to claim 5, wherein said depth gauge comprises an indexing member arranged on said top face of the bottom to facilitate the centering of said seal.
 7. The depth gauge according to claim 1, wherein said seal delimits a peripheral portion of said pressure chamber.
 8. The depth gauge according to claim 1, wherein said deformable membrane is formed by a metallic disk or by an amorphous metal alloy disk.
 9. A timepiece comprising a depth gauge comprising: a case closed by a bottom, said bottom delimiting a pressure chamber in fluidic communication with an outside of the depth gauge, a pressure sensor comprising a deformable membrane disposed above said pressure chamber; and a seal including a radially inner flank and a radially outer flank, the radially inner flank having a height greater than a height of the radially outer flank in an un-deformed state, said seal being sandwiched between the deformable membrane and the bottom to ensure tightness of said case in relation to said pressure chamber including when the deformable membrane is in a deformed state, wherein the bottom and the deformable member are devoid of a channel or a groove for housing said seal so that said seal is not retained in any channel or groove.
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